

(26 Excellent)

(19)

1. What is the primary purpose of a communication system?

- a) To transmit information signals through a communication channel. ✓
- b) To amplify signals for long-distance transmission.
- c) To reduce noise and interference in a signal.
- d) To convert electrical signals back into messages.

2. In FM broadcasting, limited coverage areas result from:

- a) Line-of-sight propagation ✓
- b) Low-frequency operation
- c) Ionospheric reflections (AM)
- d) Small bandwidth

3. The bandwidth of an SSB-SC signal is:

- a) f_c
- b) f_m ✓
- c) $2f_m$
- d) $f_c + f_m$

4. Key advantage of FM over AM is:

- a) Smaller bandwidth
- b) Simpler receivers
- c) Immunity to amplitude noise ✓
- d) Longer propagation via ionosphere

5. An AM wave is described by the equation $s(t) = A_c[1 + \mu \cos(2\pi f_m t)] \cos(2\pi f_c t)$. This equation represents which type of AM?

- a) Vestigial Side Band (VSB)
- b) Single Side Band (SSB)
- c) Double Side Band Full Carrier (DSB-FC) ✓
- d) Double Side Band Suppressed Carrier (DSB-SC)

6. The Phase Modulated signal is given by:

- a) $s(t) = A_c \cos(2\pi f_c t + \beta \cos(2\pi f_m t))$ ✓
- b) $s(t) = A_m \cos(2\pi f_m t + \beta \cos(2\pi f_c t))$
- c) $s(t) = A_c (1 + \mu m(t)) \cos(2\pi f_c t)$
- d) $s(t) = A_c \cos(2\pi f_c t) + A_m \cos(2\pi f_m t)$

7. Distortion differs from noise in that distortion:

- a) is random and unpredictable ✓
- b) Only affects digital signals
- c) is caused by external sources ✓
- d) Come from Imperfect channel response

8. In Frequency Division Multiplexing (FDM), what is used to prevent interference between different signals?

- a) Amplifiers
- b) Bandpass filters
- c) Guard bands ✓
- d) Demodulators

9. Single tone amplitude modulation

- a) contains no frequency component
- b) consists of only one frequency component ✓
- c) contains a large number of frequency components
- d) contains infinite number of frequency components

10. In FM modulation, The ratio of the maximum frequency deviation to that of frequency message, is known as:

- A) frequency interval
- B) modulation index ✓
- C) frequency deviation
- D) carrier swing

Test 1

11. In narrow band FM signal, the maximum frequency deviation is:

- A) 15 KHZ ✓ B) 51 KHZ C) 75 KHZ D) 57 KHZ

12. The imaginary line that follows the peaks of the modulated carrier wave and has the shape of the original message signal is called the:

- a) Envelope ✓ b) Sideband c) Carrier d) Broadband

13. What does the term "vestigial sideband" refer to in VSB-SC modulation?

- A) A part of the suppressed sideband is transmitted ✓

B) The complete elimination of one sideband ✗

C) The carrier signal. ✗

D) The full transmission of both sidebands ✗

14. What defines an "ideal filter" according to the provided material?

a) It amplifies signals within the passband and attenuates outside. ✗

b) It completely rejects signals within a specific band and passes all others. ~

c) It has very sharp cutoff characteristics, passing signals exactly within a specified band. ✓

d) It allows limited transmission of signals within a broad range. ✗

15. AM broadcast station transmits modulating frequency up to 6KHz. If transmitting frequency is 810KHz, then maximum and lower sidebands are _____

A. 826KHz and 804KHz

B. 916KHz and 904KHz

C. 822KHz and 816KHz

D. 816KHz and 804KHz ✓

$$USB = f_c + f_m = 810 + 6 = 816$$

$$LSB = f_c - f_m = 810 - 6 = 804$$

16. In amplitude modulation, frequency and phase of carrier:

A. varies simultaneously

B. varies alternately

C. remains constant ✓

D. Initially varies but become same after time

17. The two primary components of an Envelope Detector, as described in the block diagram, are:

a) A Summer and a Diode.

b) A Diode and a Low Pass Filter. ✓

c) A Diode and a Band Pass Filter.

d) A Local Oscillator and an AM Modulator.

18. The bandwidth of the VSBSC modulated wave:

A. $2f_m$

b. $2f_v$

c. $2f_m + f_v$

d. $f_m + f_v$ ✓

19. Which of the following is a disadvantage of SSB modulation?

A) It requires complex generation and detection processes ✓

B) It consumes more bandwidth than AM. ✗

C) It has lower power efficiency. ✗

D) It is less reliable in mobile communications. ✗

20. Which devices did we use for AM full carrier Demodulation

A) Envelope detector and Square law demodulator ✓ B. coherent

C. Ratio detector and Slope detector

D. Costas level

SOLVE THE PROBLEMA base band message : $m(t) = \cos(3000\pi t) + \cos(9000\pi t)$ And carrier signal $c(t) = 20 \cos(300000\pi t)$

the VSB shaping filter has the following design :

- transmit full USB $\rightarrow$
 - transmit vestige of LSB where $f_v = 2$ KHZ (vestige BW)
- FIND : 1. mathematical expression of DSB-SC before filter. ✓
 2. Draw the spectrum of DSB-SC. ✓
 3. bandwidth of VSB. X
 4. VSB expression equation X

Remember : $\cos a \cdot \cos b = \frac{1}{2}(\cos(a-b) + \cos(a+b))$

For tone(1)

$$F_m = \frac{3000\pi}{2\pi} = 1500 \text{ Hz}$$

For tone(2)

$$F_m = \frac{9000\pi}{2\pi} = 4500 \text{ Hz}$$

$$F_c = \frac{300000\pi}{2\pi} = 150000 \text{ Hz}$$

$$\text{DSB-SC} = A_m A_c \cos(2\pi F_m t) \cos(2\pi F_c t)$$

For tone(1)

$$\text{DSB-SC} = 20 \cos(2\pi 1500 t) \cos(2\pi 150000 t)$$

$$\text{For tone(2)} = 20 \cos(2\pi 4500 t) \cos(2\pi 150000 t)$$

$$\text{① For tone(1)} \\ 20 \cos(2\pi 150000 t) [\cos(2\pi 1500 t) \cos(2\pi 4500 t)] \\ = 20 \left[\frac{1}{2} \cos(2\pi 150000 + 1500 t) + \cos(2\pi 150000 - 1500 t) \right]$$

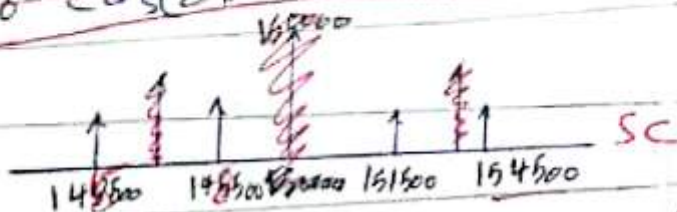
$$= 10 \cos(2\pi 151500 t) + \cos(2\pi 148500 t)$$

For tone(2)

$$20 \left[\frac{1}{2} \cos(2\pi 150000 + 4500 t) + \cos(2\pi 150000 - 4500 t) \right]$$

$$= 10 \cos(2\pi 154500 t) + \cos(2\pi 145500 t)$$

②



③

$$B_w = F_m - F_v$$

$$\Rightarrow 4500 - 2000 = 2500 \text{ Hz}$$

$$B_w = F_m + F_v$$

$$\text{④ } 10 \times \frac{F_m}{F_v} = 10 \times \frac{1}{2} = 5 \text{ V}$$

NAME: A

REG. NO.:

21.5

• SELECT THE CORRECT ANSWER : (10 MARKS)

- What is the maximum efficiency of a standard AM system with a sinusoidal message?
A) 25% (B) 33.3% C) 50% D) 100%
- What is the primary disadvantage of SSB modulation?
A) Huge bandwidth B) High power consumption
(C) Complex filter design and receiver synchronization D) Poor audio quality
- Which of the following is an advantage of PPM over PAM?
(A) Constant transmission power B) Narrower bandwidth
C) Simple receiver circuit D) No synchronization needed
- The time interval between two successive samples is called:
A) Nyquist Rate (B) Sampling Period (T_s) C) Holding Time D) Frame Rate
- The "Quadrature Null Effect" is a potential issue found in which type of detector?
A) Envelope Detector B) Square Law Detector (C) Coherent Detector D) Rectifier Detector
- A device that converts a physical message (like voice) into an electrical signal is called an:
A) Output Transducer (B) Input Transducer (C) Oscillator D) Modulator
- To recover the message from a PPM signal, it is usually first converted into:
A) AM (B) PWM C) PAM D) FM
- To prevent "Aliasing" in a sampled signal, the sampling frequency (f_s) must be:
A) $f_s < 2f_m$ (B) $f_s > 2f_m$ C) $f_s = f_m$ D) $f_s = 0$

ANSWER WITH YES OR KNOW AND CORRECT THE ANSWER (10 MARKS)

(8)

- Does "Synchronous Detection" require the receiver to have a local oscillator that is identical in frequency and phase to the carrier? No, ~~they~~ identical in single side band [SSB-SC]
- To convert PWM into PPM, a differentiator circuit followed by a monostable multivibrator is commonly used Yes.
- Is "Attenuation" the increase in signal strength as it travels through a channel? ~~Yes~~ No, The signal strength ~~stay the same~~ decrease after travels.
- Is the total power in an AM signal higher than the power in a DSB-SC signal for the same message? Yes
- Is "Multiplexing" the process of transmitting multiple signals over a single communication channel? Yes
- The Sampling Theorem states that the sampling frequency (f_s) must be less than twice the maximum message frequency (f_m)? ~~No, f_s greater than twice $\Rightarrow f_s > 2f_m$~~
- Pulse Width Modulation (PWM) is also known as Pulse Duration Modulation? Yes
- PAM signals are highly immune to noise because information is stored in the pulse amplitude. Yes.
- The process of converting a continuous-time signal into a discrete-time signal is known as? Modulation? No, Puls Modulation. (sampling)
- A Comparator is the central component used to generate PWM by comparing a modulating signal with a sawtooth waveform? Yes.